

BOOLEAN FUNCTIONS & BOOLEAN LAWS - 3

Unit 2 - Digital Electronics

2-VARIABLE MAXTERMS

A	B	Maxterm Expression	Maxterm Index
<u>0</u>	<u>0</u>	<u>$(A + B)$</u>	<u>M_0</u>
0	1	<u>$(A + B')$</u>	<u>M_1</u>
1	0	<u>$(A' + B)$</u>	<u>M_2</u>
1	1	<u>$(A' + B')$</u>	<u>M_3</u>

① minterm $\Rightarrow$ AND, $A=0 \Rightarrow A'$, $B=1 \Rightarrow B \Rightarrow$ SOP.

✓ Maxterm $\Rightarrow$ OR, $A=0 \Rightarrow A$, $B=1 \Rightarrow B' \Rightarrow$ POS.

CANONICAL POS FORM

- Let the Boolean function $F(A, B)$ be 0 for input combinations: M_0 and M_3
- So, $F(A, B) = M_0 \cdot M_3 \leftarrow \text{POS}$
- The canonical POS form is: $F(A, B) = \Pi M(0, 3) = (A+B) \cdot (A'+B') \leftarrow$

Write canonical POS for 2 variable minterms: m_1, m_2, m_3 .

3-VARIABLE MAXTERMS

	A	B	C	<u>Maxterm</u>	Decimal Index
1	0	0	0	$(A + B + C)$	M_0
2	0	0	1	$(A + B + C')$	M_1
3	0	1	0	$(A + B' + C)$	M_2
4	0	1	1	$(A + B' + C')$	M_3
5	1	0	0	$(A' + B + C)$	M_4
6	1	0	1	$(A' + B + C')$	M_5
7	1	1	0	$(A' + B' + C)$	M_6
8	1	1	1	$(A' + B' + C')$	M_7

CANONICAL POS FORM

- Any Boolean expression can be written as a **Product (AND) of maxterms** for which the function output is 0 (*Bcoz, Neutral value in Addition is 0*).

- Let the Boolean function **$F(A, B, C)$** be **0** for input combinations:

$M_0, M_2, M_4, M_6 \Rightarrow SOP \Rightarrow \bar{F}(A, B, C) = m_0 \cdot m_2 \cdot m_4 \cdot m_6$

- The canonical POS form is:

$$F(A, B, C) = \Pi M(0, 2, 4, 6) = (A+B+C) \cdot (A+B'+C) \cdot (A'+B+C) \cdot (A'+B'+C)$$

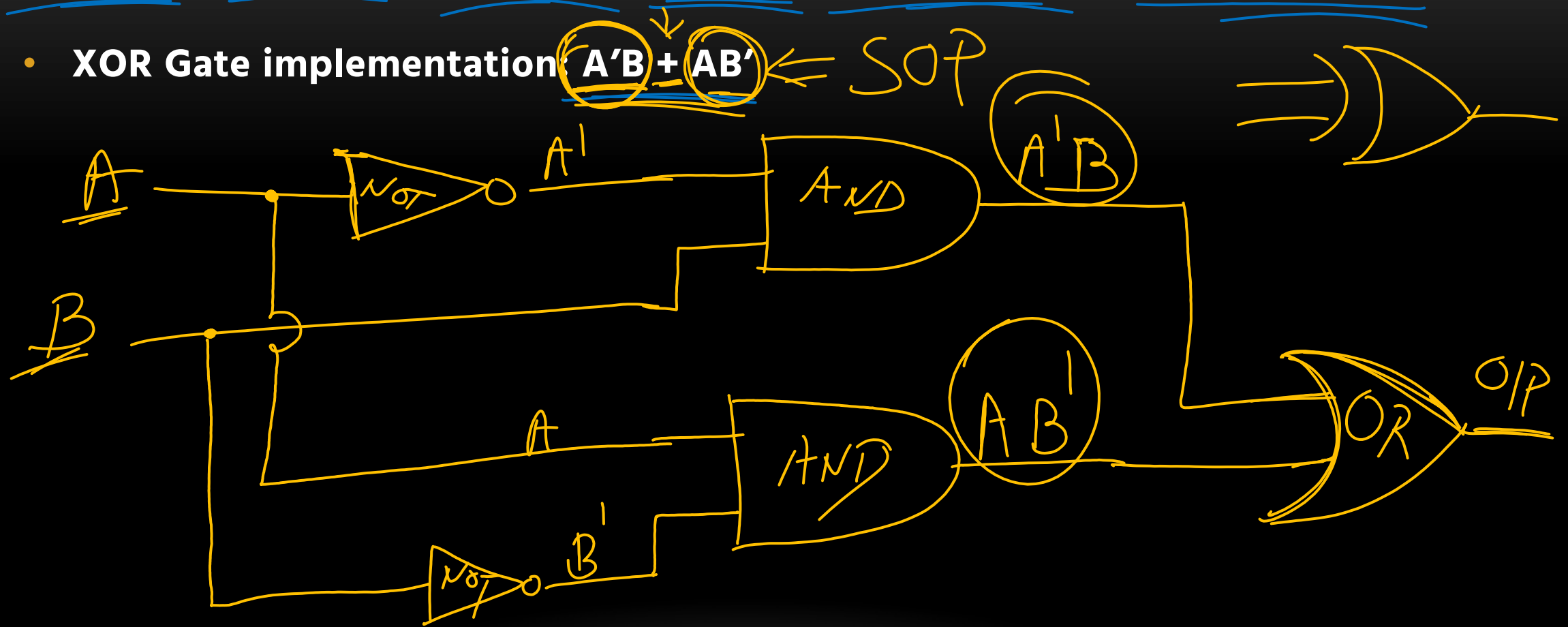
Write canonical POS for 3 variable function for: m_1, m_2, m_5, m_7

CANONICAL POS RELATIONSHIP WITH CANONICAL SOP

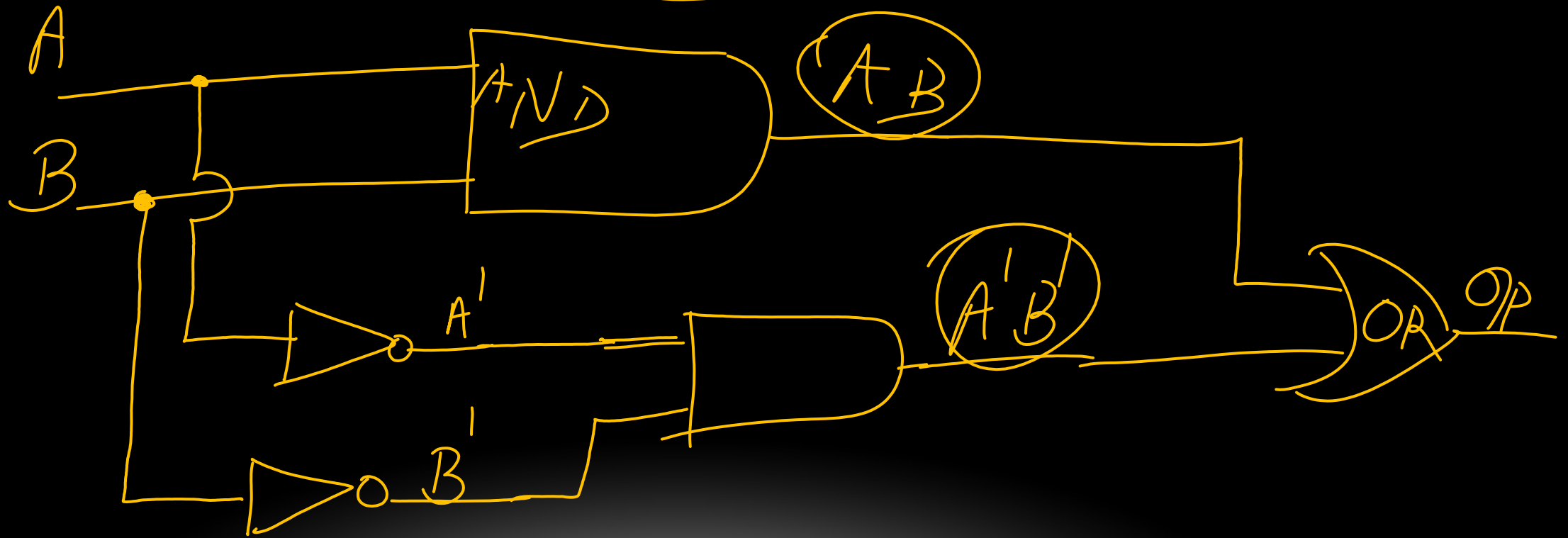
- ✓ Canonical SOP (Σm) lists minterms where $F = 1$.
- ✓ Canonical POS (ΠM) lists maxterms where $F = 0$.
- ✓ Both forms represent the same Boolean function.

DRAW LOGIC CIRCUIT FROM BOOLEAN EXPRESSION

- XOR Gate implementation: $A'B + AB'$ $\leftarrow$ SOP

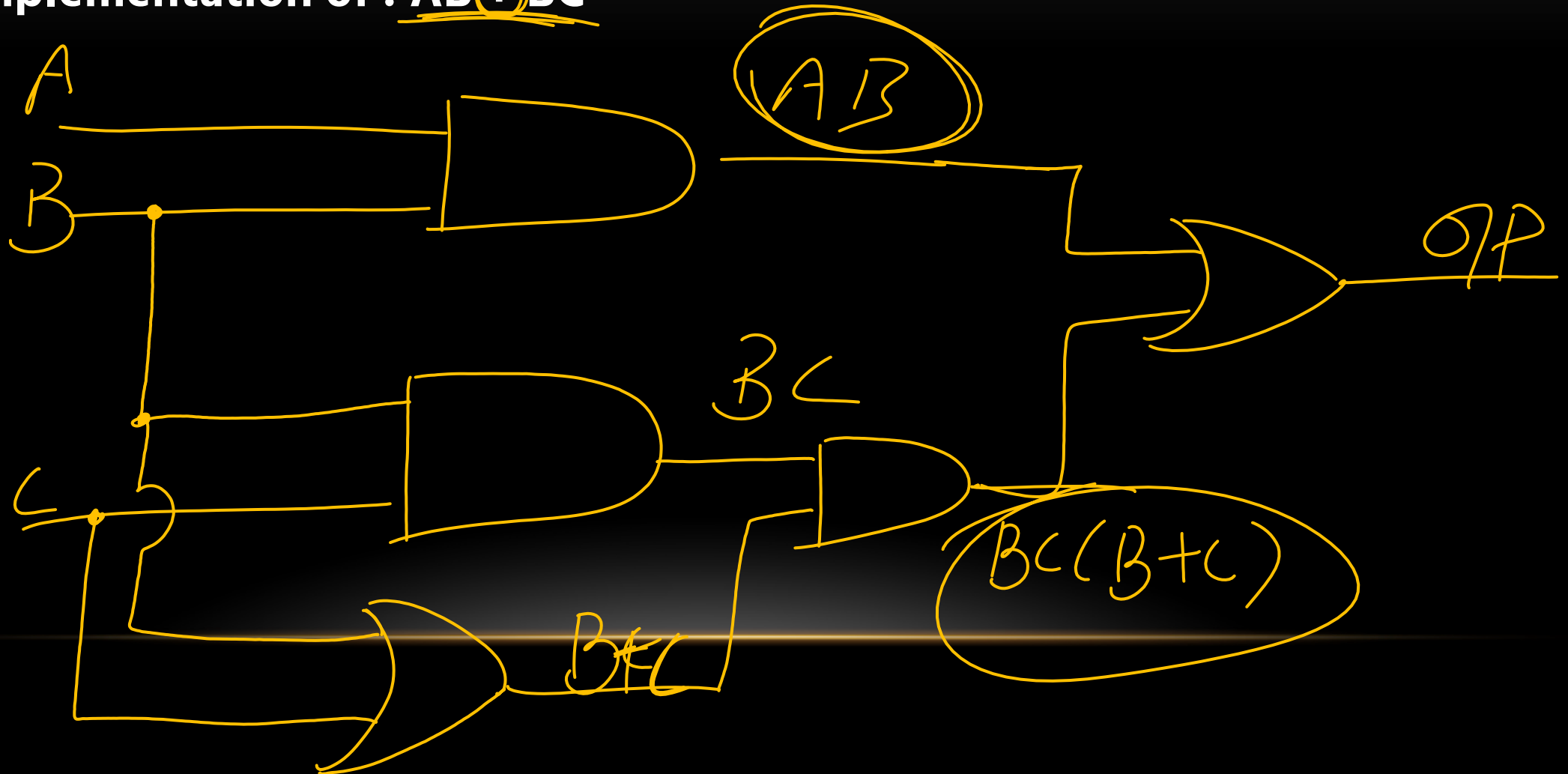


- XNOR Gate implementation: $AB + A'B'$

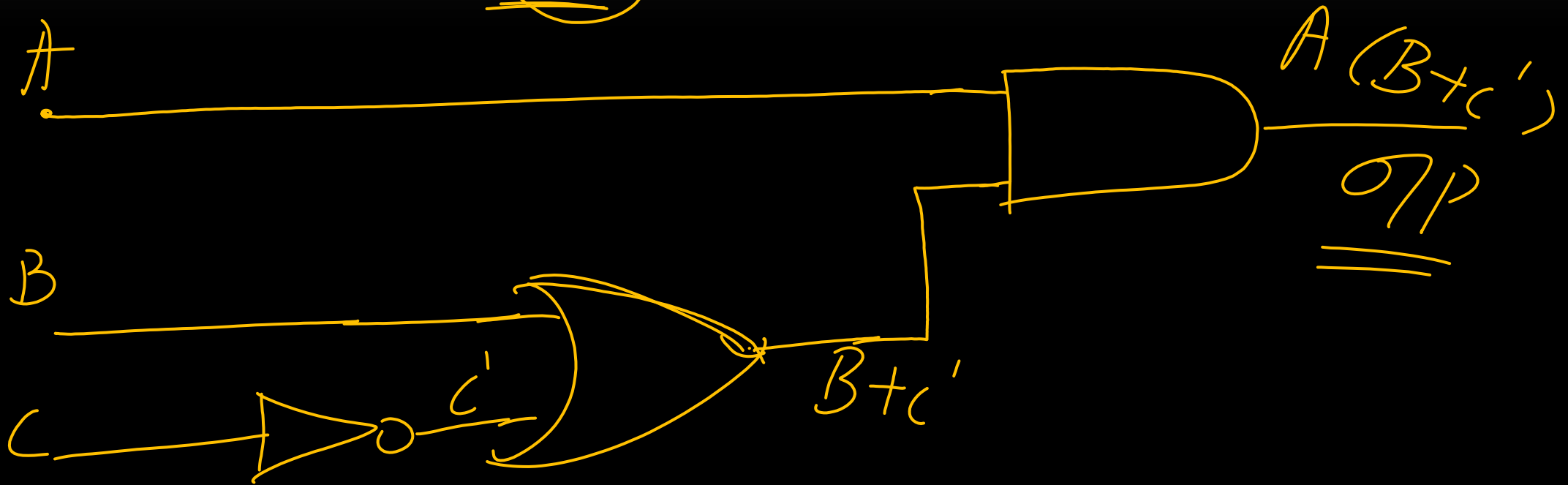


$$(AB) + BC \quad (B+C)$$

- Implementation of : $AB + BC$

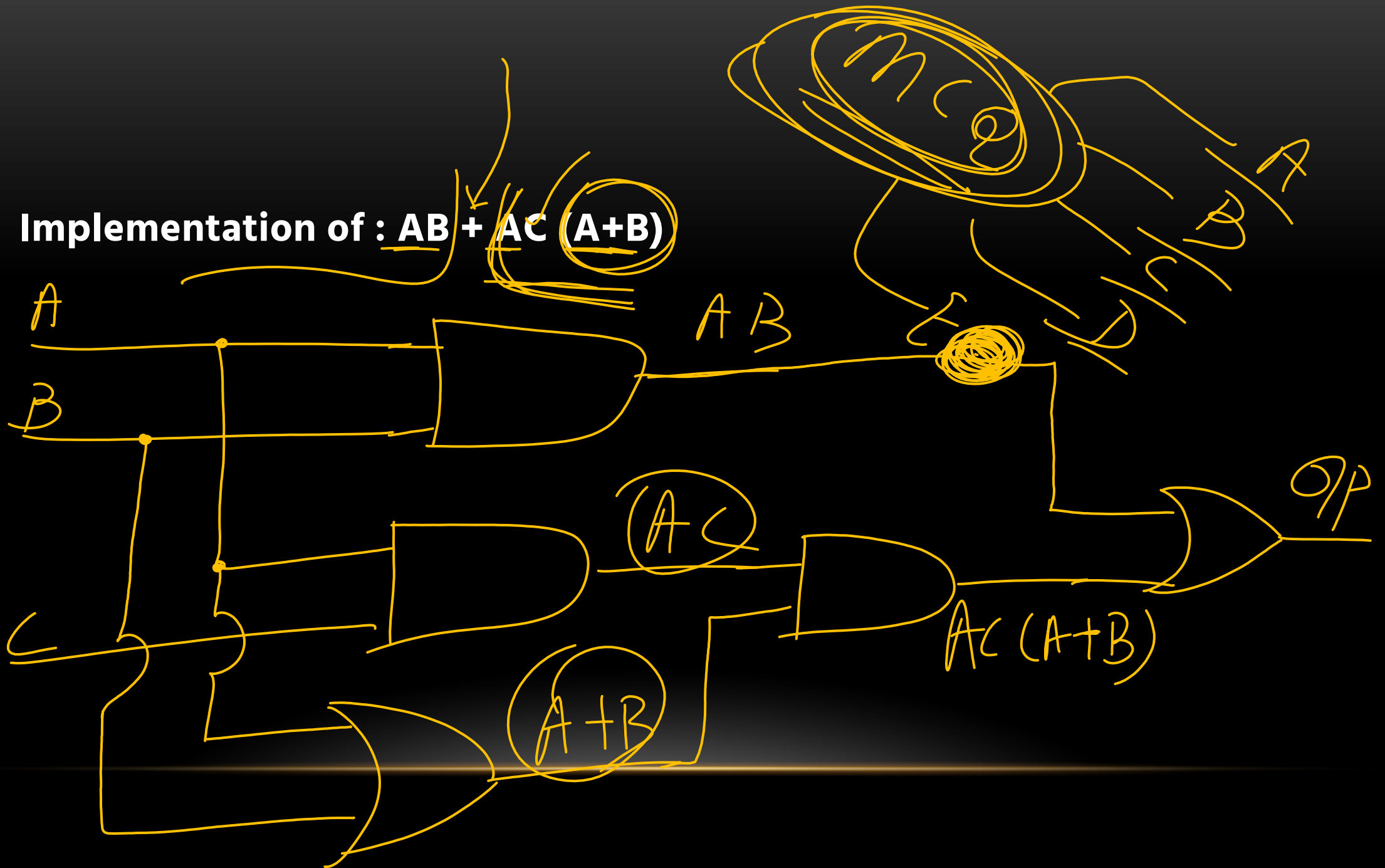


- Implementation of : $A(B+C')$



- Implementation of : $AB + BC(B+C)$

- Implementation of : $\underline{AB} + \underline{AC} (A+B)$



Proof using Boolean Laws

Find the complement of $AB + CD = 0$.

Proof: Here, $AB + CD = 0$

Applying complement both side:

$$(AB + CD)' = 0'$$

$$\therefore (AB + CD)' = 1 \quad (\because \text{Complement Law})$$

Now applying De Morgan's Theorem to the left-hand side:

$$(AB)'(CD)' = 1$$

Again applying De Morgan's Theorem:

$$(A' + B') \cdot (C' + D') = 1$$

This is the complement of the given expression.

Handwritten notes: $(AB)' = A'B'$ and $(A+B)' = A'B'$

$$\cancel{(A+B)}' = \cancel{A \cdot B}'$$

$$(A \cdot B)' = A' + B'$$

complemented of.
Rakesh Bhosker
GyanLive $\Rightarrow$ YT Lectures

Find the complement of $(A+B) \cdot (C+D) = 0$

Proof:

Take complement on both sides

$$[(A+B) \cdot (C+D)]' = (0)'$$

$$\therefore [(A+B) \cdot (C+D)]' = 1 \quad (\because \text{Complement Law})$$

Apply De Morgan's Theorem

$$(A+B)' + (C+D)' = 1$$

Again applying De Morgan's Theorem:

$$A' \cdot B' + C' \cdot D' = 1$$

This is the complement of the given expression.

Handwritten notes in yellow:
 $(A' \cdot B') + (C' \cdot D') = 1$
↑
A's } B's }
C's } D's }
अथवा
A & B }
C & D }

Handwritten note in yellow:
 $\alpha \Rightarrow \text{BOOK}$

Handwritten note in yellow:
Theory + mca $\Rightarrow$ BOOK

H/W

- Find the complement of $A + B \cdot C = 1$.

$$\Rightarrow A + (B \cdot C) = 1$$

$$\Rightarrow \boxed{A' \cdot (B' + C') = 0}$$

- Find the complement of $A' \cdot B + C = 0$.

Prove

- Find the complement of $X \cdot Y \cdot Z = 1$.

We have: $(A' \cdot B) + C = 0$

$$\Rightarrow \boxed{(A + B') \cdot C' = 1}$$

$$\begin{aligned} X' + Y' + Z' \\ = 0 \end{aligned}$$

PROOF OF FIRST ABSORPTION LAW: $A + A \cdot B = A$

Proof:

$$\underline{\text{LHS}} = \underline{A + A \cdot B}$$

$$\therefore A + A \cdot B = A \cdot (1 + B) \quad (\because \text{Distributive Law})$$

$$\therefore \text{So, } \underline{A \cdot (1 + B)} = \underline{A \cdot 1} \quad (\because \text{Identity Law})$$

$$\therefore \underline{A \cdot 1} = \underline{A} \quad (\because \text{Identity Law})$$

= RHS

Hence proved: $A + A \cdot B = A$

$$\Rightarrow x + 1 = 1$$

Proof of Second Absorption Law: $A \cdot (A+B) = A$

Proof:

$$\begin{aligned} & \dots \underbrace{(A + A)}_{\text{?}} \dots \\ & \underbrace{\hookrightarrow A} \\ & \hookrightarrow A + B \\ & \hookrightarrow B \\ & \hookrightarrow A \cdot B \end{aligned} \qquad \begin{aligned} & \underbrace{(A+B)'} = \underbrace{A' \cdot B'} \\ & \frac{(A+B)'}{A' \cdot B'} = \underbrace{1} \end{aligned}$$

PROOF OF FIRST DE MORGAN'S FIRST LAW: $(A+B)' = A'B'$

Proof: To prove equality, we show that $A'B'$ satisfies the definition of the complement of $(A+B)$.

Using the Complement Axiom: $X + X' = 1$ and $X \cdot X' = 0$

Let's Check OR condition: $(A+B) + (A'B')$

$$= [(A + B) + A'] \cdot [(A + B) + B'] \quad (\text{Using Distributive Law})$$

$$= (A + A' + B) \cdot (A + B + B') \quad (\text{Using Associative and Commutative Laws})$$

$$\text{Since } A + A' = 1 \text{ and } B + B' = 1 \quad (\text{Complement Law}):$$

$$= (1 + B) \cdot (A + 1) = 1 \cdot 1 \quad (\text{Using Commutative and Annulment law})$$

$$= 1 \cdot 1 = 1 \quad (\text{Using Axiom})$$

$$X + X' = 1$$

is correct

To Check AND Condition: $(A+B) \cdot (A'B')$

$$(AA' + BA') \cdot B'$$

$$= (0 + BA') \cdot B'$$

$$= B \cdot A' \cdot B'$$

$$= (B \cdot B') \cdot A'$$

$$= 0 \cdot A'$$

$$= 0$$

(Using Distributive Law)

(Using Complement Law: $AA' = 0$)

(Using Identity Law)

(Using Commutative and Associative Laws)

(Using Complement Law: $BB' = 0$)

(Using Annulment Law)

~~$(A+B) \cdot (A'B')$~~